Longitudinal Causal Evaluation of Mid-Stay ICU Interventions and Complications on 30-Day Unfavorable Outcomes in Patients with Pneumonia

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Background

Artificial intelligence in critical care typically relates admission characteristics to discharge outcomes, forgetting the interventions and complications that accumulate daily across a long ICU stay. Such models rank patients by risk but do not say what could be done differently [1].

Ventilator-associated and hospital-acquired pneumonia are strongly associated with mortality in ventilated patients [1], yet association cannot distinguish a marker of severity from a cause.

Why ordinary regression fails. Yesterday's ventilator settings alter today's organ failure, which in turn drives today's settings. Conditioning on daily covariates blocks the causal pathway; omitting them leaves confounding. A complication appearing on day 9 also requires nine days of survival, and discharge alive removes a patient from risk [2]. No single regression resolves these, so we use g-methods.

Objectives

To estimate the daily causal effect of mid-stay complications and of ventilator management on the probability of a 30-day unfavorable outcome, defined as in-hospital mortality or hospice discharge.

To establish which of eleven daily exposures carry effects that survive a doubly robust re-estimation, and which depend on modelling assumptions.

Research Design

Data. SCRIPT CarpeDiem v1.9.1 (PhysioNet, credentialed access): 21,931 patient-hospital-days from 704 mechanically ventilated patients who underwent bronchoalveolar lavage (BAL) for suspected pneumonia, with 955 pneumonia episodes adjudicated by a panel of ICU physicians [1].

Design. Time-zero was hospital admission, with daily follow-up to a minimum of 30 days length of stay, giving 15,428 person-days. The primary outcome occurred in 207 of 704 patients (29.4%). Discharge alive was modelled as a competing event, so 30-day status is observed for everyone.

Confounders. Age, sex, race, BMI, smoking history, Charlson index and immunocompromised status at baseline; SOFA and sub-scores, PaO_2/FiO_2, laboratory markers, vital signs, sedation depth and stay indices entered lagged one day.

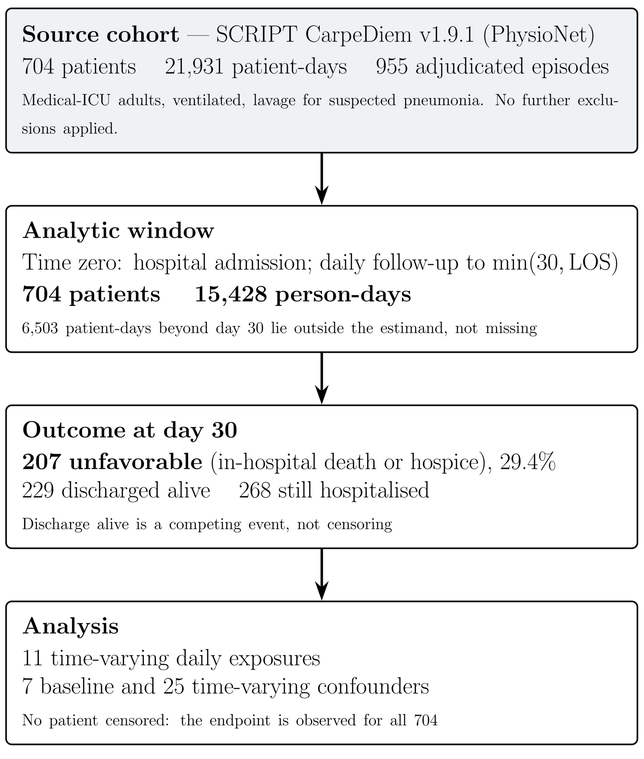


Figure 1. Cohort construction and the 30-day analytic window.

Exposure specification. Because a pneumonia episode persists once it begins, modelling it as a daily on/off exposure rendered the treatment model near-deterministic (c-statistic 0.966) and the weights uninformative; we instead modelled the hazard of episode initiation among patients not yet exposed (c-statistic 0.801).

Reported estimands. The risk difference (always versus never exposed) and the attributable risk (observed course versus never exposed), the latter being decision relevant.

Research Design (continued)

Estimation. A marginal structural model (MSM) used stabilised inverse probability of treatment weights with weighted pooled-logistic hazard models for the outcome and for discharge alive; a competing-risk g-formula recovered 30-day counterfactual risk, with cluster bootstrap inference over patients. Longitudinal targeted maximum likelihood estimation (LTMLE) was then applied by sequential regression with a logistic fluctuation step, giving doubly robust estimates [3]. Both used a discrete Super Learner, cross-fitted with folds grouped on patient.

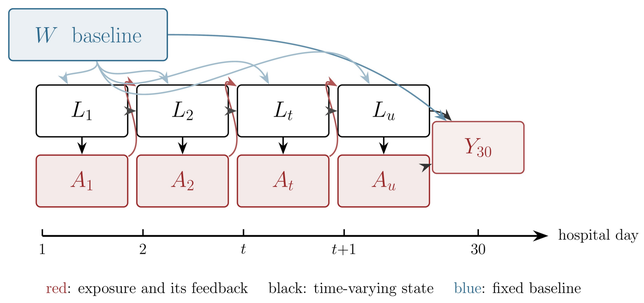


Figure 2. The daily clinical state confounds that day's exposure, which then feeds forward into the later state.

$$SW_i(t) = \prod_{k=1}^{t} \frac{\mathrm{P}\big(A_{ik} = a_{ik} \mid \bar{A}_{i,k-1},\, W_i\big)}{\mathrm{P}\big(A_{ik} = a_{ik} \mid \bar{A}_{i,k-1},\, \bar{L}_{ik},\, W_i\big)}$$

$$\widehat{\psi}(\bar{a}) = \sum_{t=1}^{30} \lambda_t^Y(\bar{a})\, S_{t-1}(\bar{a}), \qquad S_t(\bar{a}) = \prod_{k=1}^{t} \big[1 - \lambda_k^Y(\bar{a})\big]\big[1 - \lambda_k^D(\bar{a})\big]$$

Stabilised IPT weights (above); competing-risk g-formula (below).

Results and Evaluation

Ventilator management carried the identifiable effect: sustaining driving pressure ≥ 15 cmH_2O raised 30-day risk from 0.237 to 0.429, and eliminating it would lower the observed risk by 5.8 pp, approximately 41 of 704 patients. CRRT, systemic corticosteroid and FiO_2 ≥ 60% also cleared zero. Mid-stay pneumonia episodes (451 patients, 64.1%), PEEP, ECMO, antibiotics and both pathogen findings did not.

Daily exposure	Never	Always	RD, pp (95% CI)	AR, pp (95% CI)
Driving pressure ≥ 15	0.237	0.429	+19.2 (+7.4, +30.0)	+5.8 (+2.4, +8.7)
FiO_2 ≥ 60%	0.240	0.505	+26.5 (−5.9, +68.6)	+3.3 (+0.3, +8.8)
Systemic corticosteroid	0.258	0.389	+13.1 (+1.6, +25.3)	+3.7 (+0.5, +7.0)
PEEP ≥ 10	0.248	0.300	+5.2 (−13.4, +22.7)	+1.4 (−3.8, +4.5)
Mid-stay pneumonia	0.281	0.318	+3.6 (−8.3, +15.3)	+1.3 (−2.5, +4.8)
ECMO	0.276	0.529	+25.3 (−8.6, +48.7)	+1.8 (−0.2, +3.4)
CRRT	0.274	0.504	+23.0 (+11.8, +48.5)	+2.0 (+1.1, +3.6)
Antibiotic administered	0.271	0.341	+7.0 (−3.8, +22.1)	+4.5 (−1.5, +11.7)
Bronchoalveolar lavage	0.178	0.994	+81.6 (+7.8, +88.8)	+13.4 (+3.1, +19.2)
Bacterial pathogen	0.289	0.392	+10.4 (−11.6, +25.0)	+1.3 (−1.7, +3.1)
Resistant pathogen	0.292	0.455	+16.3 (−14.2, +49.4)	+0.3 (−0.4, +1.3)

Table 1. Marginal structural model with competing-risk g-formula. Observed 30-day risk 0.294.

Daily exposure	Never	Always	RD, pp (95% CI)	OR (95% CI)
Driving pressure ≥ 15	0.363	0.457	+9.4 (−7.1, +25.9)	1.48 (0.74, 2.93)
FiO_2 ≥ 60%	0.410	0.398	−1.3 (−43.2, +40.7)	0.95 (0.17, 5.45)
Systemic corticosteroid	0.381	0.408	+2.8 (−19.5, +25.0)	1.12 (0.44, 2.86)
PEEP ≥ 10	0.402	0.387	−1.5 (−23.6, +20.6)	0.94 (0.37, 2.37)
Mid-stay pneumonia	0.375	0.390	+1.6 (−12.2, +15.4)	1.07 (0.60, 1.91)
ECMO	0.406	0.394	−1.2 (−26.9, +24.5)	0.95 (0.32, 2.79)
CRRT	0.409	0.411	+0.1 (−12.3, +12.6)	1.01 (0.60, 1.69)
Antibiotic administered	0.395	0.403	+0.8 (−20.1, +21.7)	1.03 (0.43, 2.48)
Bronchoalveolar lavage	0.386	0.417	+3.2 (−27.2, +33.6)	1.14 (0.32, 4.03)
Bacterial pathogen	0.402	0.377	−2.6 (−16.7, +11.6)	0.90 (0.49, 1.64)
Resistant pathogen	0.395	0.395	−0.1 (−23.1, +23.0)	1.00 (0.38, 2.61)

Table 2. Longitudinal TMLE over a bounded ten-day window, outcome at day 30. OR is the marginal odds ratio.

What worked as expected. For exposures that genuinely vary day to day, the two estimators agree: +9.1 against +9.4 pp for driving pressure, +1.9 against +1.6 for complications. Natural-course calibration reproduced the observed 30-day risk (0.295 against 0.294).

What did not, and why. For procedures whose always-exposed regime never occurs in practice, the MSM extrapolates: daily lavage gives +76.7 pp against the LTMLE's +3.2. That is a positivity violation made visible, not an effect. CRRT and corticosteroid cleared zero under the MSM but not under LTMLE, and both are escalated in deteriorating patients, so confounding by indication is the expected reading.

Results: Model Agreement

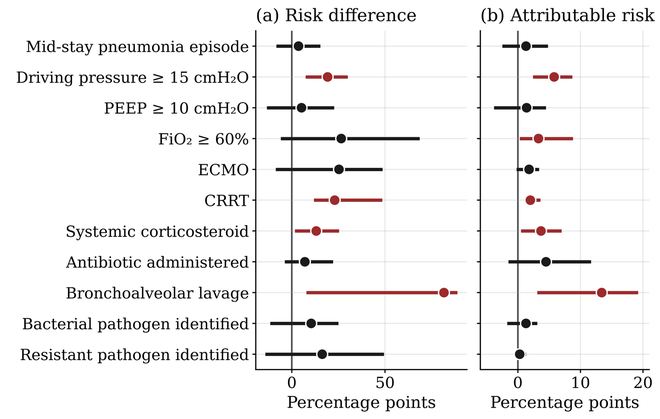


Figure 3. MSM estimates for all eleven exposures; intervals excluding zero in red.

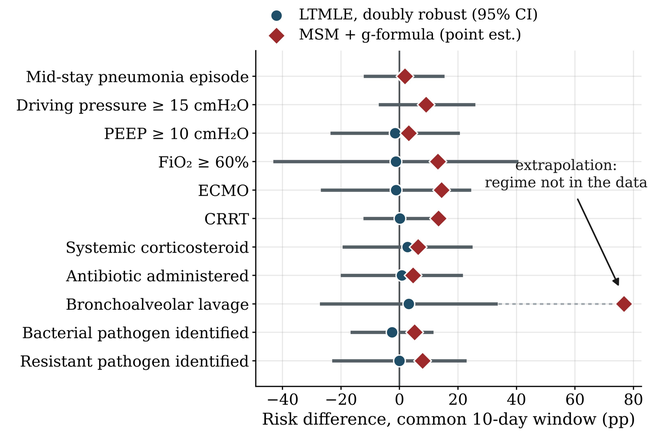


Figure 4. Both estimators on one bounded estimand; a long connector marks a regime with no support in the data.

Conclusions

The ventilator, not the infection. Secondary pneumonia contributes little to 30-day risk once immortal time and treatment-confounder feedback are handled, whereas a modifiable ventilator parameter carries a substantial attributable risk that both estimators recover [4,5].

Most exposures carried no identifiable effect. PEEP, ECMO, antibiotics and the bacterial and resistant pathogen findings were compatible with no effect under both estimators.

Diagnostics. E-values 3.02 (driving pressure) and 1.51 (complications); randomisation inference consistent with the null for complications ($p = 0.50$); admission-day prediction AUROC 0.638.

Limitations. A single-centre cohort, a disposition-based endpoint, and the assumption of no unmeasured confounding at each time point.

Future Plans

Separate episode occurrence from episode resolution, testing whether the harm lies in acquiring a second pneumonia or in failing to clear one.

Replace static always/never regimes with modified treatment policies, and treat ventilator settings as continuous rather than thresholded.

Replicate externally in multi-centre data (MIMIC-IV, eICU-CRD) and extend follow-up beyond hospital discharge.

References

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Data: SCRIPT CarpeDiem v1.9.1, PhysioNet credentialed access (doi.org/10.13026/n9xd-6d71). No patient-level data are reproduced here.
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